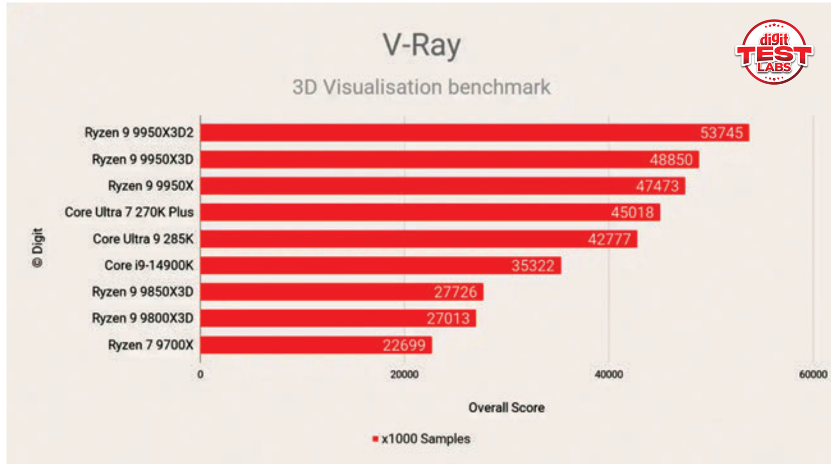
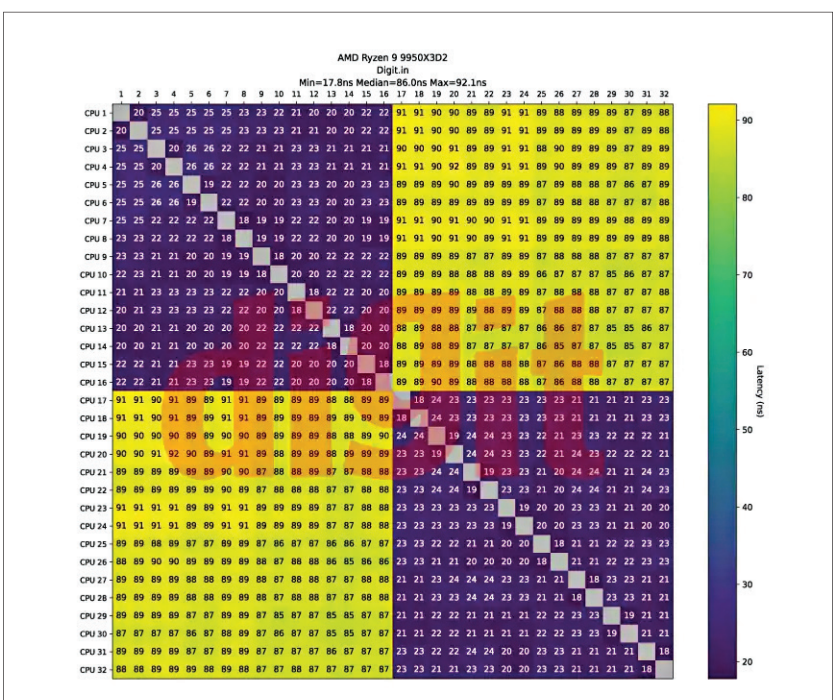
is slight drop compared to the 9950X3D. The Median latency was 81 nanoseconds with the 9950X3D and that's increased to 86 nanoseconds. It's still an improvement over the older gen processors wherein the communication between CCDs would have a larger penalty.



THERMALS AND POWER

For such a cache-heavy, high-power desktop CPU, the Ryzen 9 9950X3D2 appears surprisingly well-behaved in this configuration. The recorded average CPU package temperature of 36 degrees Celsius suggests that the NZXT 360 mm cooler is doing its job comfortably under sustained mixed load conditions. Even the peak temperature of 90 degrees Celsius remains well within AMD's specified thermal ceiling of 95 degrees Celsius, leaving some headroom rather than pushing the silicon to the brink.

That matters, because this is a 200 W-class processor designed for serious all-core work, not just short gaming bursts. Power draw tells the more revealing story. An average package power of 230 W and a peak of 250 W make it clear that this chip is happy to consume substantial power when fully unleashed, which tracks with AMD's 270 W PPT figure. In practical terms, the 9950X3D2 is not pretending to be an efficiency-first flagship.



VERDICT

The Ryzen 9 9950X3D2 looks like AMD taking the X3D idea to its logical extreme on AM5. By putting second-generation 3D V-Cache on both CCDs, AMD has created a part that is clearly aimed at buyers who want more than just top-tier gaming. This processor is meant for serious desktop users working across rendering, compiling, simulation, AI-adjacent workloads, and content creation, while still retaining the strong gaming DNA associated with the X3D family. On paper, the gains over the existing 9950X3D are not universal, but the expanded cache and more aggressive power envelope do give it a more specialised and more ambitious identity.

Its 200 W TDP and high package power figures mean it belongs in an uncompromising desktop build, not in a value-focused or modestly cooled setup. The slight dip in maximum boost clock versus the regular 9950X3D also makes it clear that AMD is prioritising cache-rich workload behaviour over headline frequency. Taken as a whole, it feels less like a mainstream recommendation and more like a statement product. For the right workload mix, it could be one of AM5's most interesting processors yet.

–Mithun Mohandas

SPECIFICATIONS

Architecture: Zen 5 | Socket: AM% | Cores: 16 | Threads: 32 | Base Clock: 4.3 GHz | Boost clock: 5.6 GHz | Cache: 208 MB | 3D V-Cache design: 2nd Gen | TDP: 200 W | PPT: 270 W | EDC: 250 A | Max temp: 95 C | Memory support: DDR5

CONTACT

AMD | Telephone: NA | Email: support.amd.com | Website: www.amd.com

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